A GENERIC MEMBER OF A SUFFICIENTLY AMPLE LINEAR SYSTEM STAYS GEOMETRICALLY IRREDUCIBLE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9054-9071 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 9073-9159, the complete author proof (85 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The author's statement contains 'in reference {\cite[Theorem 6.3 part 4)]{Jou}}', so the card carries a bibliography fragment. That fragment reproduces, field for field and without editing, the @BOOK{Jou} entry from the Stacks Project snapshot bibliography shipped with this corpus at focused-seeds/AGP/sources/stacks-snapshot-20261011/my.bib (same upstream Stacks Project snapshot). No entry text was altered and no citation was reconstructed from memory.

Editorial note. A three-line wrapper preamble span supplies the mechanical reference/slogan/history environments, which the retained chapter preamble does not define; this follows the pattern already used by other Stacks cards in this corpus.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Ample line bundles and their linear systems, geometric irreducibility, and generic flatness; the cited inputs are prerequisites and the generic-member argument is retained.

Source reference. Original lemma lemma-algebraically-independent: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9012–9022. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-amazing-bertini-lemma: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8907–8910. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-geom-irreducible-generic-fibre: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7557–7564. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma varieties-lemma-geometrically-irreducible-function-field: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9159–9159. Cited by the retained statement or proof but not part of the collected spans;

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supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma varieties-situation-family-divisors: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9159–9159. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. [1, Theorem 6.3 part 4)] *In Varieties, Situation varieties-situation-family-divisors assume*

(1) *X is of finite type over k,*
(2) *X is geometrically irreducible over k,*
(3) *there exist $v_1, v_2, v_3 \in V$ and an irreducible component Z of $H_{v_2} \cap H_{v_3}$ such that $Z \not\subset H_{v_1}$ and $\text{codim}(Z, X) = 2$, and*
(4) *every irreducible component Y of $\bigcap_{v \in V} H_v$ has $\text{codim}(Y, X) \geq 2$.*

Then for general $v \in V \otimes_k k'$ the scheme H_v is geometrically irreducible over k'.

Proof. In order for assumption (3) to hold, the elements v_1, v_2, v_3 must be k-linearly independent in V (small detail omitted). Thus we may choose a basis $v_1, \ldots, v_r$ of V incorporating these elements as the first 3. Recall that $H_{univ} \subset \mathbf{A}^r_k \times_k X$ is the "universal divisor". Consider the projection $q : H_{univ} \to \mathbf{A}^r_k$ whose scheme theoretic fibres are the divisors H_v. By Lemma lemma-geom-irreducible-generic-fibre it suffices to show that the generic fibre of q is geometrically irreducible. To prove this we may replace X by its reduction, hence we may assume X is an integral scheme of finite type over k.

Let $U \subset X$ be a nonempty affine open such that $\mathcal{L}|_U \cong \mathcal{O}_U$. Write $U = \text{Spec}(A)$. Denote $f_i \in A$ the element corresponding to section $\psi(v_i)|_U$ via the isomorphism $\mathcal{L}|_U \cong \mathcal{O}_U$. Then $H_{univ} \cap (\mathbf{A}^r_k \times_k U)$ is given by

$$H_U = \text{Spec}(A[x_1, \ldots, x_r]/(x_1 f_1 + \ldots + x_r f_r))$$

By our choice of basis we see that f_1 cannot be zero because this would mean $v_1 = 0$ and hence $H_{v_1} = X$ which contradicts assumption (3). Hence $\sum x_i f_i$ is a nonzerodivisor in $A[x_1, \ldots, x_r]$. It follows that every irreducible component of H_U has dimension $d + r - 1$ where $d = \dim(X) = \dim(A)$. If $U' = U \cap D(f_1)$ then we see that

$$H_{U'} = \text{Spec}(A_{f_1}[x_1, \ldots, x_r]/(x_1 f_1 + \ldots + x_r f_r)) \cong \text{Spec}(A_{f_1}[x_2, \ldots x_r]) = \mathbf{A}^{r-1}_k \times_k U'$$

is irreducible. On the other hand, we have

$$H_U \setminus H_{U'} = \text{Spec}(A/(f_1)[x_1, \ldots, x_r]/(x_2 f_2 + \ldots + x_r f_r))$$

which has dimension at most $d + r - 2$. Namely, for $i \neq 1$ the scheme $(H_U \setminus H_{U'}) \times_U D(f_i)$ is either empty (if $f_i = 0$) or by the same argument as above isomorphic to an $r - 1$ dimensional affine space over an open of $\text{Spec}(A/(f_1))$ and hence has dimension at most $d + r - 2$. On the other hand, $(H_U \setminus H_{U'}) \times_U V(f_2, \ldots, f_r)$ is an r dimensional affine space over $\text{Spec}(A/(f_1, \ldots, f_r))$ and hence assumption (4) tells us this has dimension at most $d + r - 2$. We conclude that H_U is irreducible for every U as above. It follows that H_{univ} is irreducible.

Thus it suffices to show that the generic point of H_{univ} is geometrically irreducible over the generic point of $\mathbf{A}^r_k$, see Varieties, Lemma varieties-lemma-geometrically-irreducible-function-field. Choose a nonempty affine open $U = \text{Spec}(A)$ of X contained in $X \setminus H_{v_1}$ which meets the irreducible component Z of $H_{v_2} \cap H_{v_3}$ whose

existence is asserted in assumption (3). With notation as above we have to prove that the field extension

$$\mathrm{Frac}(A[x_1, \ldots, x_r]/(x_1 f_1 + \ldots + x_r f_r))/k(x_1, \ldots, x_r)$$

is geometrically irreducible. Observe that f_1 is invertible in A by our choice of U. Set $K = \mathrm{Frac}(A)$ equal to the fraction field of A. Eliminating the variable x_1 as above, we find that we have to show that the field extension

$$K(x_2, \ldots, x_r)/k(x_2, \ldots, x_r, -\sum\nolimits_{i=2,\ldots,r} f_1^{-1} f_i x_i)$$

is geometrically irreducible. By Lemma lemma-amazing-bertini-lemma it suffices to show that for some $c_2, \ldots, c_r \in k$ the elements

$$x_2, \ldots, x_r, \sum\nolimits_{i=2,\ldots,r} f_1^{-1} f_i x_i, \sum\nolimits_{i=2,\ldots,r} c_i f_1^{-1} f_i$$

are algebraically independent over k in the fraction field of $A[x_2, \ldots, x_r]$. This follows from Lemma lemma-algebraically-independent and the fact that $Z \cap U$ is an irreducible component of $V(f_1^{-1} f_2, f_1^{-1} f_3) \subset U$. □

References

[1] Jouanolou, Jean-Pierre. *Théorèmes de Bertini et Applications.* Progress in Mathematics 42. Birkhäuser, 1983.